

## PROBLEM FROM PREVIOUS LECTURE

Testing hypothesis on the mean of normal with unknown  $\sigma^2$ .

**Ex:** # of medical in-patient days in nursing homes.

Sample of  $n = 18$  normal random variables, unknown  $\mu$  and  $\sigma^2$ .

$\bar{X}_n = 182.17$ ,  $\hat{\sigma}_1 = 72.22$ .      Test  $H_0: \mu \geq 200$ ,  $H_A: \mu < 200$

Already seen  $\frac{\sqrt{n}(\bar{X}_n - \mu_0)}{\hat{\sigma}_1}$  has *t-distribution* with  $n - 1$  dof.

Observed value:  $\frac{\sqrt{18}(182.7 - 200)}{72.22} = -1.018$

## PROBLEM FROM PREVIOUS LECTURE

Testing hypothesis on the mean of normal with unknown  $\sigma^2$ .

**Ex:** # of medical in-patient days in nursing homes.

Sample of  $n = 18$  normal random variables, unknown  $\mu$  and  $\sigma^2$ .

$\bar{X}_n = 182.17$ ,  $\hat{\sigma}_1 = 72.22$ .      Test  $H_0: \mu \geq 200$ ,  $H_A: \mu < 200$

**Alternative-1** to present results:

```
from scipy.stats import t
t.ppf(0.05, 17)
```

-1.7396067260750676

Since observed value =  $-1.018 > -1.73$  we don't reject  $H_0$ .

# HYPOTHESIS TESTING

Testing hypothesis on the mean of normal with unknown  $\sigma^2$ .

**Ex:** # of medical in-patient days in nursing homes.

Sample of  $n = 18$  normal random variables, unknown  $\mu$  and  $\sigma^2$ .

$\bar{X}_n = 182.17$ ,  $\hat{\sigma}_1 = 72.22$ .      Test  $H_0: \mu \geq 200$ ,  $H_A: \mu < 200$

**Alternative-2** to present results:



# HYPOTHESIS TESTING

Testing hypothesis on the mean of normal with unknown  $\sigma^2$ .

**Ex:** # of medical in-patient days in nursing homes.

Sample of  $n = 18$  normal random variables, unknown  $\mu$  and  $\sigma^2$ .

$\bar{X}_n = 182.17$ ,  $\hat{\sigma}_1 = 72.22$ .      Test  $H_0: \mu \geq 200$ ,  $H_A: \mu < 200$

**Alternative-2** to present results:

```
from scipy.stats import t  
t.cdf(-1.018, 17)
```

0.16147397056756546

Since p-value =  $0.16 > 0.05$  we don't reject  $H_0$ .

# HYPOTHESIS TESTING

## Comparing sample means

Measure the arm lengths of two groups. Does it show one group has longer arms, on average? **Is the difference significant?**

# HYPOTHESIS TESTING

## Comparing sample means

Measure the arm lengths of two groups. Does it show one group has longer arms, on average? **Is the difference significant?**

**Two sample t-test:** Are means of two samples different significantly?

2-sided example:

$H_0$ : Same means       $H_1$ : different means



# HYPOTHESIS TESTING

Comparing sample means    Unknown but common variance  $\sigma^2$

# HYPOTHESIS TESTING

Comparing sample means Unknown but common variance  $\sigma^2$

The two sample t test:

$X = (X_1, \dots, X_m)$  random sample from normal,  $\mu_1, \sigma^2$  unknown

$Y = (Y_1, \dots, Y_n)$  random sample from normal,  $\mu_2, \sigma^2$  unknown



# HYPOTHESIS TESTING

Comparing sample means    Unknown but common variance  $\sigma^2$

Test hypotheses:  $H_0: \mu_1 \leq \mu_2$ ,  $H_1: \mu_1 > \mu_2$

# HYPOTHESIS TESTING

Comparing sample means    Unknown but common variance  $\sigma^2$

Test hypotheses:  $H_0: \mu_1 \leq \mu_2$ ,  $H_1: \mu_1 > \mu_2$

**Theorem:**  $T = \frac{\sqrt{m+n-2}(\bar{X}_m - \bar{Y}_n)}{\sqrt{\frac{1}{m} + \frac{1}{n}} \sqrt{\sum_{i=1}^m (X_i - \bar{X}_m)^2 + \sum_{i=1}^n (Y_i - \bar{Y}_n)^2}}$  has *t distribution* with  $m+n-2$  dof.

At significance  $\alpha$ , reject  $H_0$  if  $T \geq F_v^{-1}(1 - \alpha)$ , for  $v = m + n - 2$  dof.

Alternatively present p value too:  $1 - F_v(T)$ , for  $v = m + n - 2$  dof.

# HYPOTHESIS TESTING

Comparing sample means    Unknown variances  $\sigma_1^2, \sigma_2^2$  different.



# HYPOTHESIS TESTING

Comparing sample means    Unknown variances  $\sigma_1^2, \sigma_2^2$  different.

Use Welch's t statistic.

# HYPOTHESIS TESTING

Comparing sample means    Unknown variances  $\sigma_1^2, \sigma_2^2$  different.

Use Welch's t statistic.

Ex: Effect of a calcium supplement on blood pressure.

10 people received calcium supplement. 11 received placebo.

|         |    |    |    |    |    |    |   |    |     |    |    |
|---------|----|----|----|----|----|----|---|----|-----|----|----|
| Calcium | 7  | −4 | 18 | 17 | −3 | −5 | 1 | 10 | 11  | −2 |    |
| Placebo | −1 | 12 | −1 | −3 | 3  | −5 | 5 | 2  | −11 | −1 | −3 |

Each entry:  
Blood pressure after 12 weeks —  
Blood pressure before experiment

# HYPOTHESIS TESTING

Comparing sample means    Unknown variances  $\sigma_1^2, \sigma_2^2$  different.

Use Welch's t statistic.

Ex: Effect of a calcium supplement on blood pressure.

10 people received calcium supplement. 11 received placebo.

|         |    |    |    |    |    |    |   |    |     |    |                                                                                    |  |
|---------|----|----|----|----|----|----|---|----|-----|----|------------------------------------------------------------------------------------|--|
| Calcium | 7  | -4 | 18 | 17 | -3 | -5 | 1 | 10 | 11  | -2 | Each entry:<br>Blood pressure after 12 weeks —<br>Blood pressure before experiment |  |
| Placebo | -1 | 12 | -1 | -3 | 3  | -5 | 5 | 2  | -11 | -1 |                                                                                    |  |

Does the calcium supplement increase blood pressure?

$\mu_1$ : mean change in blood pressure of calcium supplement group

$\mu_2$ : mean change in blood pressure of placebo group

$H_0: \mu_1 \leq \mu_2, \quad H_A: \mu_1 > \mu_2$       Test significance at  $\alpha = 0.05$ .



# HYPOTHESIS TESTING

## Calcium supplement group vs placebo

```
import scipy.stats as stats
import numpy as np

calcium_sample = np.array([7, -4, 18, 17, -3, -5, 1,
                           10, 11, -2])
np.mean(calcium_sample)
```

5.0

```
np.var(calcium_sample)
```

68.8

```
placebo_sample = np.array([-1, 12, -1, -3, 3, -5, 5,
                           2, -11, -1, -3])
np.mean(placebo_sample)
```

-0.2727272727272727

```
np.var(placebo_sample)
```

31.65289256198347

```
stats.ttest_ind(calcium_sample, placebo_sample, equal_var = False, alternative = 'greater')
```

TtestResult(statistic=1.6037172876755148, pvalue=0.06441968481096698, df=15.590512968733774)



# HYPOTHESIS TESTING

## Calcium supplement group vs placebo

```
import scipy.stats as stats
import numpy as np

calcium_sample = np.array([7, -4, 18, 17, -3, -5, 1,
                           10, 11, -2])
np.mean(calcium_sample)
```

5.0

```
np.var(calcium_sample)
```

68.8

```
placebo_sample = np.array([-1, 12, -1, -3, 3, -5, 5,
                           2, -11, -1, -3])
np.mean(placebo_sample)
```

-0.2727272727272727

```
np.var(placebo_sample)
```

31.65289256198347

**Note 1:** For Welch's t test, set to False

```
stats.ttest_ind(calcium_sample, placebo_sample, equal_var = False, alternative = 'greater')
```

TtestResult(statistic=1.6037172876755148, pvalue=0.06441968481096698, df=15.590512968733774)



# HYPOTHESIS TESTING

## Calcium supplement group vs placebo

```
import scipy.stats as stats
import numpy as np

calcium_sample = np.array([7, -4, 18, 17, -3, -5, 1,
                           10, 11, -2])
np.mean(calcium_sample)
```

5.0

```
np.var(calcium_sample)
```

68.8

```
placebo_sample = np.array([-1, 12, -1, -3, 3, -5, 5,
                           2, -11, -1, -3])
np.mean(placebo_sample)
```

-0.2727272727272727

```
np.var(placebo_sample)
```

31.65289256198347

Note 2: When  $H_A$  is  $\mu_1 > \mu_2$ .

```
stats.ttest_ind(calcium_sample, placebo_sample, equal_var = False, alternative = 'greater')
```

TtestResult(statistic=1.6037172876755148, pvalue=0.06441968481096698, df=15.590512968733774)

# HYPOTHESIS TESTING

## Calcium supplement group vs placebo

```
import scipy.stats as stats
import numpy as np

calcium_sample = np.array([7, -4, 18, 17, -3, -5, 1,
                           10, 11, -2])
np.mean(calcium_sample)
```

5.0

```
np.var(calcium_sample)
```

68.8

```
placebo_sample = np.array([-1, 12, -1, -3, 3, -5, 5,
                           2, -11, -1, -3])
np.mean(placebo_sample)
```

-0.2727272727272727

```
np.var(placebo_sample)
```

31.65289256198347

Can not reject  $H_0$  at 0.05 significance level

```
stats.ttest_ind(calcium_sample, placebo_sample, equal_var = False, alternative = 'greater')
```

```
TtestResult(statistic=1.6037172876755148, pvalue=0.06441968481096698, df=15.590512968733774)
```

# THE $\chi^2$ TEST

Given explanatory variable and response variable (both categorical)  
test for association between the variables.

$H_0$ : No association between variables

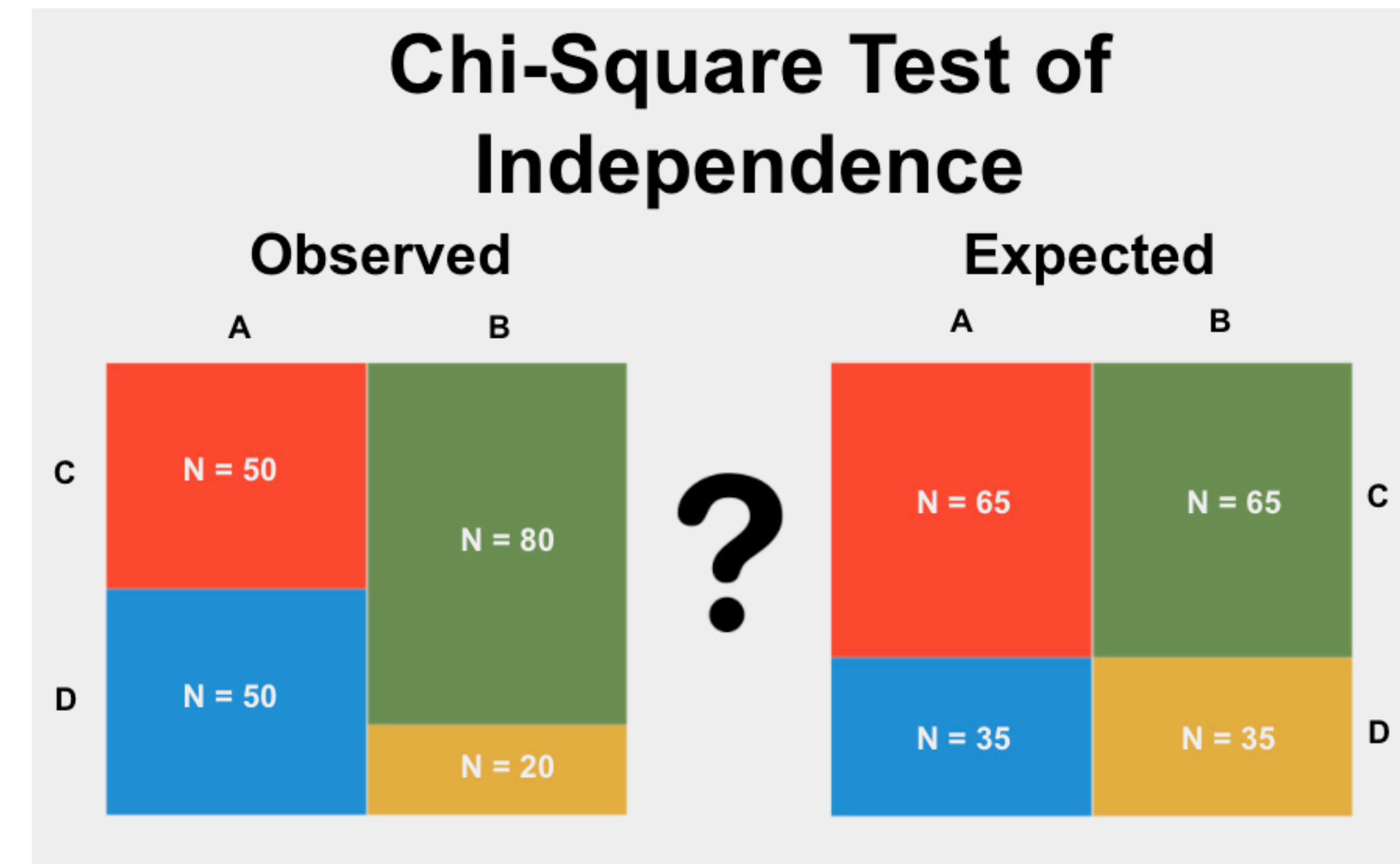


# THE $\chi^2$ TEST

Given explanatory variable and response variable (both categorical)  
test for association between the variables.

$H_0$ : No association between variables

- Build contingency table
- Find out expected values
- Compute p value of  $\chi^2$  statistic
- If p value significant  
    reject  $H_0$  and claim association  
else  
    not enough evidence for association





# THE $\chi^2$ TEST

**Ex:** Is cancer independent of marital status?

| Marital Status | Cancer      | No Cancer   | Total |
|----------------|-------------|-------------|-------|
| Single         | 29 (38.1)   | 47 (37.9)   | 76    |
| Married        | 116 (112.3) | 108 (111.7) | 224   |
| Widowed        | 67 (61.6)   | 56 (61.4)   | 123   |
| Div/Sep        | 5 (5.0)     | 5 (5.0)     | 10    |
| Total          | 217         | 216         | 433   |

How to find expected values?  
(Green shaded values)

Assuming independence:  
(row sum x column sum) / total

# THE $\chi^2$ TEST

Ex: Is cancer independent of marital status?

| Marital Status | Cancer      | No Cancer   | Total |
|----------------|-------------|-------------|-------|
| Single         | 29 (38.1)   | 47 (37.9)   | 76    |
| Married        | 116 (112.3) | 108 (111.7) | 224   |
| Widowed        | 67 (61.6)   | 56 (61.4)   | 123   |
| Div/Sep        | 5 (5.0)     | 5 (5.0)     | 10    |
| Total          | 217         | 216         | 433   |

How to find expected values?  
(Green shaded values)

Assuming independence:  
(row sum x column sum) / total

How to compute  $\chi^2$  statistics?

$$X^2 = \sum \frac{(\text{observed} - \text{expected})^2}{\text{expected}} = \frac{(29 - 38.1)^2}{38.1} + \frac{(47 - 37.9)^2}{37.9} + \dots + \frac{(5 - 5.0)^2}{5.0} = 5.53$$

# THE $\chi^2$ TEST

**Ex:** Is cancer independent of marital status?

| Marital Status | Cancer      | No Cancer   | Total |
|----------------|-------------|-------------|-------|
| Single         | 29 (38.1)   | 47 (37.9)   | 76    |
| Married        | 116 (112.3) | 108 (111.7) | 224   |
| Widowed        | 67 (61.6)   | 56 (61.4)   | 123   |
| Div/Sep        | 5 (5.0)     | 5 (5.0)     | 10    |
| Total          | 217         | 216         | 433   |

How to find expected values?  
(Green shaded values)

Assuming independence:  
(row sum x column sum) / total

How to compute  $\chi^2$  statistics?

$$X^2 = \sum \frac{(\text{observed} - \text{expected})^2}{\text{expected}} = \frac{(29 - 38.1)^2}{38.1} + \frac{(47 - 37.9)^2}{37.9} + \dots + \frac{(5 - 5.0)^2}{5.0} = 5.53$$

```
from scipy.stats import chi2
1-chi2.cdf(5.53, 3)
```

```
0.13685525539157362
```



# THE $\chi^2$ TEST

**Ex:** Is cancer independent of marital status?

| Marital Status | Cancer      | No Cancer   | Total |
|----------------|-------------|-------------|-------|
| Single         | 29 (38.1)   | 47 (37.9)   | 76    |
| Married        | 116 (112.3) | 108 (111.7) | 224   |
| Widowed        | 67 (61.6)   | 56 (61.4)   | 123   |
| Div/Sep        | 5 (5.0)     | 5 (5.0)     | 10    |
| Total          | 217         | 216         | 433   |

How to find expected values?  
(Green shaded values)

Assuming independence:  
(row sum x column sum) / total

How to compute  $\chi^2$  statistics?

$$X^2 = \sum \frac{(\text{observed} - \text{expected})^2}{\text{expected}} = \frac{(29 - 38.1)^2}{38.1} + \frac{(47 - 37.9)^2}{37.9} + \dots + \frac{(5 - 5.0)^2}{5.0} = 5.53$$

```
from scipy.stats import chi2
1-chi2.cdf(5.53, 3)
```

0.13685525539157362

$\text{dof} = (\text{row\_count} - 1)(\text{column\_count} - 1)$

# THE $\chi^2$ TEST

**Ex:** Is cancer independent of marital status?

| Marital Status | Cancer      | No Cancer   | Total |
|----------------|-------------|-------------|-------|
| Single         | 29 (38.1)   | 47 (37.9)   | 76    |
| Married        | 116 (112.3) | 108 (111.7) | 224   |
| Widowed        | 67 (61.6)   | 56 (61.4)   | 123   |
| Div/Sep        | 5 (5.0)     | 5 (5.0)     | 10    |
| Total          | 217         | 216         | 433   |

How to find expected values?  
(Green shaded values)

Assuming independence:  
(row sum x column sum) / total

How to compute  $\chi^2$  statistics?

$$X^2 = \sum \frac{(\text{observed} - \text{expected})^2}{\text{expected}} = \frac{(29 - 38.1)^2}{38.1} + \frac{(47 - 37.9)^2}{37.9} + \dots + \frac{(5 - 5.0)^2}{5.0} = 5.53$$

```
from scipy.stats import chi2
1-chi2.cdf(5.53, 3)
```

0.13685525539157362

p value of test statistic: 0.137

At significance level 0.05 won't reject  $H_0$

# THE $\chi^2$ TEST

**Ex:** Is cancer independent of marital status?

| Marital Status | Cancer      | No Cancer   | Total |
|----------------|-------------|-------------|-------|
| Single         | 29 (38.1)   | 47 (37.9)   | 76    |
| Married        | 116 (112.3) | 108 (111.7) | 224   |
| Widowed        | 67 (61.6)   | 56 (61.4)   | 123   |
| Div/Sep        | 5 (5.0)     | 5 (5.0)     | 10    |
| Total          | 217         | 216         | 433   |

How to find expected values?  
(Green shaded values)

Assuming independence:  
(row sum x column sum) / total

How to compute  $\chi^2$  statistics?

$$X^2 = \sum \frac{(\text{observed} - \text{expected})^2}{\text{expected}} = \frac{(29 - 38.1)^2}{38.1} + \frac{(47 - 37.9)^2}{37.9} + \dots + \frac{(5 - 5.0)^2}{5.0} = 5.53$$

```
from scipy.stats import chi2
1-chi2.cdf(5.53, 3)
```

0.13685525539157362

p value of test statistic: 0.137

At significance level 0.05 won't reject  $H_0$

⇒ Can't conclude marital status associated with cancer.